

Assessment Ref. No.						Activity Assessed					
Assessment Date						Faculty / Directorate					
Assessor						School / Service					
Version No.						Additional individuals involved in developing the RA					
Signature of Assessor						Signature of Academic Supervisor / Approver					
Risk Score Matrix							Risk Score and Description				
Severity							Risk Score	Risk Level	Category	Description	
Likelihood		Insignificant	Minor	Moderate	Major	Fatal					
	Very Unlikely	1 Green	2 Green	3 Green	4 Green	5 Amber	1 – 4	Low	Acceptable	No further actions needed	
	Unlikely	2 Green	4 Green	6 Amber	8 Amber	10 Red	5 – 9	Medium	Tolerable/Adequate	Should be reviewed to ensure that there is nothing else that can be done	
	Possible	3 Green	6 Amber	9 Amber	12 Red	15 Red	10 – 15	High	Undesirable	Immediately review current control measures, and where appropriate decide on further actions	
	Likely	4 Green	8 Amber	12 Red	16 Red	20 Red	16 - 25	Very High	Unacceptable	Stop activity and make immediate improvements	
	Almost Certain	5 Amber	10 Red	15 Red	20 Red	25 Red	<i>Likelihood (L) x Severity (S) = Risk Score (RS)</i>				

What is/are the hazard(s) involved with the activity being undertaken?	Who might be harmed and how?	What are you already doing to control the risk?	Risk Score with current controls in place			What further action is necessary? (Add these actions to the action plan below).	Target Risk Score Likelihood x Severity = Risk Score		
			L	S	RS		L	S	RS
Scope creep	The project could overrun if I were to keep adding new features to the game. This would impact the product I would present at the end of the year as if I were to	Creating the product backlog Using a gantt chart to plan out the project timeline Working using agile methodology to ensure adaptability throughout the project	3 - Possible	3 - Moderate	9 - Medium Risk	Ensure I do not take on more than what is already planned unless it is absolutely necessary Make regular updates to my gantt chart	2 - Unlikely	3 - Moderate	6 - Medium Risk

	run out of time I would be presenting an unfinished product								
Knowledge restrictions	Will impact the time the project takes causing me to overrun A worse or unfinished product could be made due to not implementing key features due to not knowing how	Learning SDL in my free time having already done multiple projects Breaking down tasks into simpler problems to not get overwhelmed	3 - Possible	4 - Major	12 - High Risk	Continue to learn SDL Seek further help if struggling whether this be colleagues or using AI	2 - Unlikely	4 - Major	8 - Medium Risk
Memory leaks	Reduce performance of my game leading to a less enjoyable experience for the player Could crash the project	Testing code during development to see if any memory leaks are occurring. Testing through both playing my game and debugging the software	4 - Likely	3 - Moderate	12 - High Risk	Utilise AI or existing software's to help test my code for memory leaks Create thorough test plans User testing	3 - Possible	3 - Moderate	9 - Medium Risk
Game bugs	Creates a worse playing experience for the users. Could lead to ruining the enjoyment of the game and impact its success	Testing code throughout development Keeping log of bugs that occur	5 - Almost Certain	3 - Moderate	15 - High Risk	Utilise AI to help test my code Unit testing User testing	5 - Almost Certain	2 - Minor	10 - High Risk
Game being inefficient	Impacts the player creating a worse game experience May mean the game cannot be ran on older or worse devices limiting the potential player base	Optimising code while developing. Getting a working version then prioritising efficiency	3 - Possible	4 - Major	12 - High Risk	Use AI to help find where my code is inefficient Debugging my code during testing	2 - Unlikely	4 - Major	8 - Medium Risk

Game not being fun	Leads to the game dying quickly and not being successful Players won't want to get a game that isn't fun	Playing the game myself throughout development Finding what makes existing Real Time Strategy (RTS) games fun and ensuring I utilise the same techniques to good effect	3 - Possible	5 - Fatal	15 - High Risk	Have others test my game frequently throughout development and gain their feedback Ensure those that test my game are from a variety of backgrounds and enjoy RTS games to ensure the feedback I am getting is varied and reliable	2 - Unlikely	5 - Fatal	10 - High Risk
Artwork not being very good	Impacts game feel and the immersion of the game leading to a worse game experience for the player	Use pixel art which I am more familiar with making Keep to a simple and consistent art style	4 - Likely	3 - Moderate	12 - High Risk	Use AI to help generate artwork for my game Use preexisting art for inspiration to help my work	2 - Unlikely	3 - Moderate	6 - Medium Risk
Maintaining hardware developing on	Impacts my motivation if my hardware being used is slow Not being able to work at home adds to the time I need to spend on the project due to hour journey each way to and from the university to use university systems	I got my one virus install of the year out of the way before the end of sprint 1 – this is a joke Maintain my PC ensure it is secure and up to date	5 - Almost Certain	4 - Major	20 - Very High Risk	Install anti-virus and get a better firewall to keep my pc secure	2 - Unlikely	3 - Moderate	6 - Medium Risk
Dependencies and Licences	Dependencies and software being used to develop my game have licences which I need to make sure I am compliant with else I am at risk of break of contract.	Install decency versions I am using so that I don't need to update them Ensure I check any change of licences of software's I am using	1 - Very Unlikely	4 - Major	4 - Low Risk	N/A	1 - Very Unlikely	4 - Major	4 - Low Risk

Loss of user data e.g. levels complete	Would force players to replay levels they have already beat. This is not particularly fun to do and would ruin the game experience.	N/A	3 - Possible	3 - Moderate	9 - Medium Risk	Ensure during testing I test the reading and writing to files Unit testing	1 - Very Unlikely	3 - Moderate	3 - Low Risk
Players gaming the system	If not stored in an appropriate way players could go into the games files and unlock levels/features they should not otherwise be able to modify	N/A	4 - Likely	2 - Minor	8 - Medium Risk	Encode data stored in files so the user cannot change them e.g. use binary files	2 - Unlikely	2 - Minor	4 - Low Risk
AI usage	During development can create bugs and issues as code generated by AI is not reliable. Although I will not be using AI generated code. When using it to help figure how to tackle a task it could lead me down the wrong path	Check AI's answers against my own knowledge as well as online help	3 - Possible	2 - Minor	6 - Medium Risk	N/A	3 - Possible	2 - Minor	6 - Medium Risk

Refer to scoring matrix and likelihood / severity descriptors

Likelihood Descriptors

Likelihood of injury / harm	Examples	Score
Very unlikely	Good control measures are in place. Controls do not rely on a person using them (i.e. personal compliance with safety rules). Controls are very unlikely to break down. People are very rarely in this area or very rarely engage in this activity.	1
Unlikely	Reasonable control measures are in place but they do rely on a person using them (some room for human error). Controls unlikely to breakdown. People are not often in this area / do not often engage in this activity.	2
Possible	Inadequate controls are in place, or likely to breakdown if not maintained. Controls rely on personal compliance. People are sometimes in this area or sometimes engage in this activity and situations sometimes arise from this activity.	3
Likely	Poor controls in place. Heavy reliance on personal compliance (lots of room for human error). People are often in this area / engage in this activity on a regular basis / situation often arise from this activity.	4
Almost certain	No controls in place where there should be, exposure to the hazard is expected to occur in most circumstances. The activity is considered such high risk that it will certainly lead to injuries.	5

Severity Descriptors

Severity of injury / harm	Examples	Score
Insignificant	None or very insignificant injuries, health effects, damage or disruption to work. Short-term and/or localised environmental harm.	1
Minor	Cuts bruises, mild skin irritations, mild headaches and pains requiring minor first aid treatment. Minor property damage or disruption to work. Notable contributor to environmental harm.	2
Moderate	More serious injuries or ill-health requiring time off work or a hospital visit for example burns sprains, strains, short term musculoskeletal disorders, cut requiring stitches, back injuries, fractures to fingers and toes. Short term absence relating to physical or mental health issues. More serious property damage or disruption. A significant contributor to environmental harm.	3
Major	Broken limbs, amputations, long-term health problems or longer absence. Acute illness requiring medical treatment. Loss of consciousness, serious electric shock, loss of sight. Major property damage, major disruption to work. A major contributor to significant environmental harm.	4
Fatal	Injury or ill-health which leads to death either at the time, soon after the incident, or eventually, as in the case of certain occupational diseases, such as asbestos-related cancers. Catastrophic business losses. The major contributor to significant environmental harm.	5